

Team LONG PULL — 2026-07-09 (2nd pull). Casey's TWO-PHASE cosmology aligns the whole board: PHASE 1 = D_{IV}^5 resonance (cosmological, clean, step-1, scale-free — the manifold “sings a new note” when a channel shuts); PHASE 2 = in-bud dynamics (matter, messy, step-2, emission). PRIMARY: THE BUILD (one build settles CP + masses). date + SOD.

Landing adjudications from pull #1 (Keeper, light touch)

- Elie's Λ CDM cluster — BANK CANDIDATES on my desk, pending Cal's independence check: $\Omega_\Lambda=13/19$ (0.07σ , banked K666), $\Omega_m=6/19$ (SAME object as Ω_Λ — not independent), $\Omega_{DM}/\Omega_b=\text{rank}^4/N_c=16/3$ (0.47σ , Wallach-shadow mechanism — STRONG, matches the corpus 16/3 at 0.2%, not form-cheap), $A_s=(3/4)\alpha^4$ (0.97σ), $n_s=1-5/137$ (0.33σ , pending cascade-forced). Independence is the gate — Ω_Λ/Ω_m are one; A_s shares α . Cal reconciles how many INDEPENDENT banks this cluster is before I stamp. η_B : existence=cant-sign (structural, forced one bit — GOOD); magnitude $2\alpha^4/(3\pi)$ at 2σ (honest lead, not bank).
- Grace's $m_s/m_d=20=\text{rank}^2 \cdot n_C$ — HELD identified-tier, NOT a clean bank (form-cheap: also $(5/3) \cdot 12$, also a Wallach dim — over-determined, no single forced mechanism). The frame is VALIDATED (methodological gain: the sector circled because we graded the intrinsically-fuzzy b/d ratio) but it does not hand a bank. The ~ 330 MeV emission dressing is NOT derived — hard open.
- Lyra's CP — honest reframe ACCEPTED: it's a COMPUTATION, not an abstract theorem. The detail-first check caught a real red herring (the $SO(2)$ /clock CHARGE is removable by global rephasing → NOT the CP source; the clock SIGN = handedness, survives). CP requires the 3 generations to be 2D-complex-spread (non-collinear as points in $\mathbb{C}$). Real risk named: if they're real-collinear → no asymmetry → CP-from- $N_c=3$ fails. Only the build settles it. Good discipline — no premature theorem.
- Keeper's step-1/step-2 lens — VALIDATED target-innocently by BOTH lanes (Grace's scale-invariant split + Elie's all-scale-free cluster). Upgraded by Casey's resonance mechanism (below): step-1 = D_{IV}^5 's own resonance response; step-2 = in-bud dynamics.

Casey's two-phase picture (the organizing frame)

D_{IV}^5 resonates when the flow is modified and an output channel shuts off — it feels a shift, displaces energy (sings a new note), and once a bud universe begins forming it settles into a new harmony. The bud is written from the first resonance; the new note begins the energy transfer (kettle → purer tone). So: cosmological parameters = D_{IV}^5 's RESPONSE (Phase 1 — pure manifold resonance, clean/geometric/step-1). Then inside the bud, dynamics start (Phase 2 — messy, matter, what's NOT tied to manifold geometry, step-2/emission). Open question Casey poses: why does information turn into matter — debris of the calculation, or a receipt?

PRIMARY — THE BUILD (Phase 2 key: one build, both answers) · LYRA + GRACE JOINT

Construct the $SO(5,2)$ oscillator / linear-algebra states, find the THREE generation states, read their peak positions. This ONE build settles: - CP (Lyra): are the 3 peaks 2D-complex-spread (non-collinear in $\mathbb{C}$ → Jarlskog $\neq 0$ → CP exists, $N_c=3$ forced) or real-collinear (→ no CP → the idea fails)? The honest fork. - Masses (Grace): the same peak positions set the mass spacing (the step-1 bare/current values, before emission dressing). - Lyra + Grace share the build — Lyra reads the complex spread, Grace reads the radial spacing. Same states, both answers. This is the highest-value move on the board: it unparks masses AND settles CP simultaneously, in BST's own scheme. Use the linear-algebra construction (remember linear algebra).

PHASE 1 mechanism (Casey's new physics) — THE RESONANCE NOTE · ELIE

The Λ CDM cluster is harvested (Phase 1 outputs). Now the MECHANISM: when an output channel of D_{IV}^5 shuts off, what “note” does it sing? Build the toy — D_{IV}^5 's spectral response to a shut channel: does the resonance shift produce a COMPUTABLE harmonic (a new note = the cosmological output — Ω_Λ , n_s as the manifold's harmonics)? If the cosmological parameters are D_{IV}^5 's resonant response, there should be a resonance/eigenvalue computation that FORCES them as harmonics, not just clean forms that fit. That would turn the Phase-1 cluster from “forced-looking topological ratios” into “the manifold's resonance spectrum” — the

mechanism under the harvest. (Also consolidate the Λ CDM cluster for Cal's null-model with the independence caveats.)

CAL — the cosmological independence reconciliation + count

How many INDEPENDENT banks is the Λ CDM cluster (Ω_Λ/Ω_m are one object; A_s shares α ; Ω_{DM}/Ω_b and n_s look independent)? Null-model the cluster honestly. Finalize the reconciled count including the confirmed independent cosmological banks.

KEEPER — adjudicate at landing

Stamp the cosmological cluster once Cal returns the independence count (don't double-count Ω_m ; watch the A_s/α shared- α). Hold Grace's m_s/m_d identified-tier. Track the build (the CP fork is a genuine falsifier — if real-collinear, CP-from- $N_c=3$ fails honestly). Adjudicate the resonance toy if it lands. Light touch, at landing.

Disciplines (light)

BST's own scheme, not λ QCD. The build is the priority — one construction, CP + masses. Cosmological cluster: independence before banking (don't inflate a correlated cluster). The resonance note: a computation that FORCES the harmonics, not a fit. Grace's m_s/m_d : identified-tier, form-cheap, honest. The CP fork is real (it can fail). Count as Cal reconciles.

— Keeper, 2026-07-09 pull #2. Casey's two-phase cosmology aligns the board: Phase 1 = D_{IV}^5 resonance (clean cosmological cluster, step-1) $\rightarrow$ Phase 2 = in-bud dynamics (matter, step-2). PRIMARY: THE BUILD (Lyra+Grace joint — $SO(5,2)$ states, 3 generation peaks; settles CP spread AND mass spacing in one construction). Elie: the resonance-note mechanism (does a shut channel force the cosmological harmonics?) + consolidate Λ CDM for Cal. Cal: cosmological independence/null-model + count. Keeper: stamp the cluster on Cal's independence check (Ω_Λ/Ω_m one object; A_s shares α ; $\Omega_{DM}/\Omega_b=16/3$ strong), hold m_s/m_d identified-tier, track the CP fork (real falsifier). Step-1/step-2 lens validated both lanes.